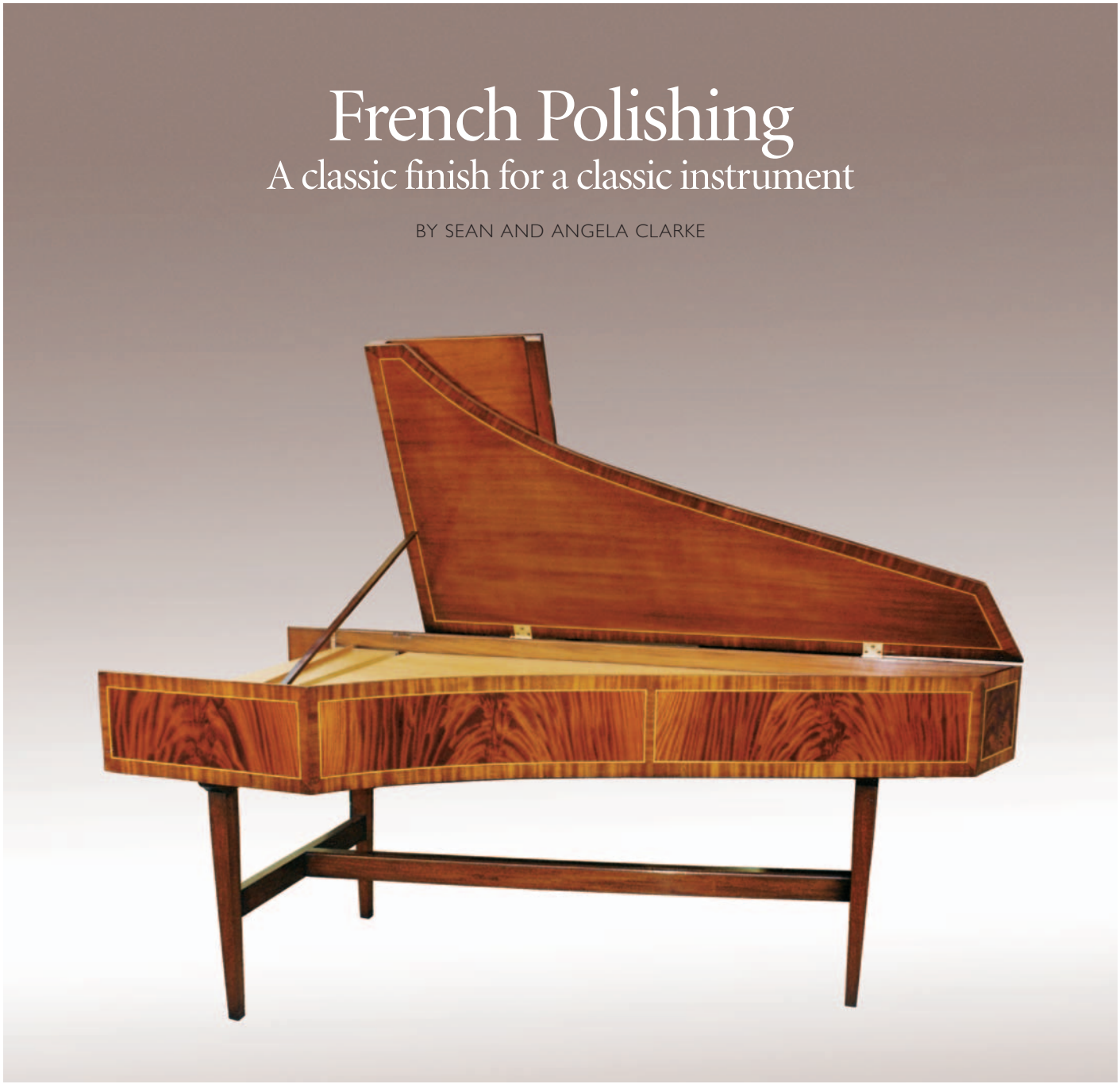
French Polishing

A classic finish for a classic instrument

BY SEAN AND ANGELA CLARKE



It's not everyday that I'm asked to work on a harpsichord, but recently a client who had built one from a kit asked me to veneer and apply a finish to it. I immediately proposed that it be French polished, since I knew that this traditional hand-rubbed finish would greatly illuminate the beautiful quartersawn and crotch mahogany veneers, making this instrument truly sing.

Developed in France around 1820, French polishing involves the hand application of layers of shellac, which penetrate deep into the wood fibers, creating depth and clarity that are unmatched. All French polishers have their own particular methods, and the one I will describe here is one that works best for me.

It consists of a few basic steps. First, the application of an initial sealing coat, known as a "fad coat" in Britain, applied along the direction of the grain. This acts as the base for further shellac build-up. Then, the application of a grain filler, which is pressed into the grain, creating a less porous and flatter surface in order to build a full-bodied finish. Next, bodying up, which is the application of the shellac build, done using a rubber [**see sidebar**] and applied in a circular and figure-eight motion. And finally, spiriting off, which is the removal of the linseed oil lubricant from the finish, and also the phase where the polishing in French polishing occurs.

After the veneer work was completed, I assessed the overall look of the piece. I liked the natural color of the crotch mahogany and decided to proceed without any staining. However, I did want to

tone down the mahogany banding and top. Since the color change I wanted was subtle, I decided to slightly tint those areas after the grain had been filled. So I went straight to the application of my first fad coat.

I generally prefer super blonde shellac for fad coats, because darker shellacs tend to leave color streaks. I use Moser's shellac in the flake form [**available at Woodworker's Supply, (800) 645-9292**]. Premixed versions are available but have a limited shelf life. Because the flakes take several hours to dissolve, I had mixed my shellac the night before in a 2 lb. cut. The "cut" refers to the ratio of shellac to its solvent, denatured alcohol. A 2 lb. cut is 2 lb. shellac per gallon of alcohol. Of course, it's possible (and usually the practice) to mix less, as long as the ratio remains the same. Although I most often work with a 2 lb. cut, I followed the harpsichord manufacturer's recommendation when it came time to do the spruce sound board, and fadded it with a 1 lb. cut of super blonde shellac.

I dipped my fad (an open-knit cotton pad) with the super blonde shellac, squeezed out the excess, and applied a thin coat in straight strokes along the grain (**1**). I wiped one coat on the entire body of the harpsichord, and brushed one coat on the harpsichord stand. Often, on smaller surface areas like legs, it's easier to brush on shellac (**2**). I repeated this process 3-4 times until there was a consistent sheen and color. This fadding up process allows the shellac to soak right into the grain, enhancing the depth of the veneer. I then let the piece dry for a few hours, lightly sanded all surfaces with 320-grit, and dusted it off (**3**).

Next, I began to apply the grain filler, darkening it and making it more visible—giving it more "pop." For this project, I used Bartley's grain filler [**available at Woodworker's Supply**], and brushed it on all surfaces, across the grain (**4**). I let it gel for about 10 minutes, and then wiped off the excess with a piece of wood I had fashioned for this purpose, again working across the grain (**5**). A plastic scraper could also work. I wiped off the piece a final time with a cotton cloth. (Some polishers like to use burlap here.) All surfaces were clean with a dull, even sheen, so I let the piece dry overnight.

















The following morning, I used #0000 steel wool and rubbed across the grain (**6**) to make sure that all surfaces were completely clear of the grain filler, which would impede the finish later. After dusting off the piece, I applied a few more fad coats in the manner previously described to seal in the grain filler (**7**). At this point, I evaluated the color of the piece, because grain filler can slightly alter color. I still felt that the mahogany banding and lid were too light, and that there should be less contrast with the crotch mahogany inlay. So I formulated a warm brown spirit color consisting of 2 cups each alcohol and super blonde shellac, 2 tsp. each black and yellow aniline dyes, and 1 tsp. red aniline dye [**dyes available at Mohawk Finishing Products, (800) 545-0047**].

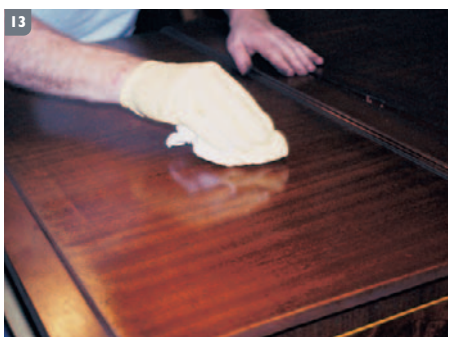
I taped off the crotch mahogany and boxwood inlays, and used a squirrel hair brush to apply the tint in small areas of about 6" square (**8**). After brushing the mixture on an area, I immediately used a piece of cotton dampened with alcohol and patted it on the color to even it out.



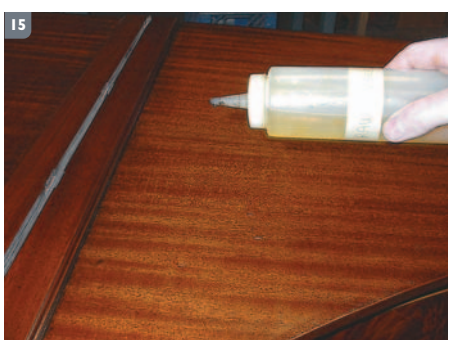


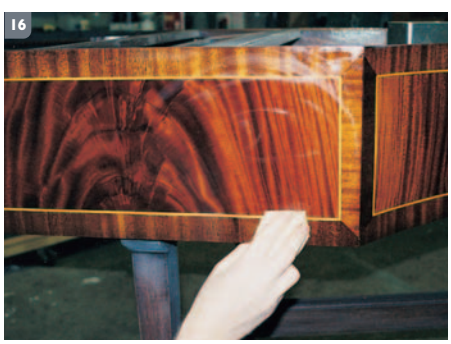




















Since the lid was a much larger surface area, I decided to save the color tint there for the bodying up stage. I allowed the newly tinted banding to dry for about one hour, then removed the tape.

The owner of the harpsichord had requested a slightly distressed finish, with some age and character, so next I physically distressed the piece with a stone (**9**). I used the stone to scratch and dent in various spots, trying to concentrate in areas where distressing would have naturally occurred. To illuminate the scratches and dents as well as slightly age the surfaces, I created a glaze to apply. My tried-and-true glaze recipe is as follows: 3 tsp. burnt umber pigment, 1 tsp. vegetable black pigment [**Myland's pigments available at Davis International, (888) 369-5263**], 1 oz. Rolco gold size [**available at Easy Leaf Products (800) 569-5323 or at local art supply stores**], and 4 oz. turpentine. I applied the glaze by brush to a single panel at a time (**10**), then used a cotton rag to wipe it off immediately (**11**). For a hand-rubbed and shaded effect, I tried to leave the glaze a little heavier on the edges and in the crooks and crannies. After all panels were dry, I started back at the first one and used #0000 steel wool to further remove the glaze (**12**), leaving the centers of the panels and the boxwood inlay clean. I let the glaze dry overnight.

The next day, I began by dusting off the entire harpsichord, followed by the



MAKING A RUBBER

First, I cut a piece of cotton (I use a good quality bed sheet) into a square about 6" x 6". This will form the outer part of the rubber, and is sometimes called a "linen." Next, I cut a piece of cotton wadding about 6" square, and fold it over several times. I moisten the wadding with denatured alcohol for absorbency, then add about 1 oz. of shellac to one surface of the wadding. I place the charged side of the wadding face down onto the center of the linen, then fold each corner of the linen into the center. I fold around one corner of the wadding, forming a point on the top of the rubber. (This point helps the rubber access corners and crevices when working.) I twist all of the linen corners together to form a "handle" for the rubber, and make sure that the flat surface on the opposite side of the rubber is completely smooth. I tap the flat surface of the rubber on the palm of my hand several times, so that the shellac in the wadding penetrates the linen. The rubber is charged and ready to go.

Note: This is only one way to create a rubber. Experiment to find what works best for you.

application of another few fad coats of shellac to seal in the glaze. This time, I used a pad to pat on the initial fad coat (**13**), since any dragging motion could pull the glaze. I let the patted-on coat dry for about one hour, then fadded on another two coats in the usual manner, in straight strokes with the grain. Another overnight dry-time was necessary.

I was then ready to begin the bodying process, where successive coats of shellac would be applied, resulting in the depth for which French polishing is famous. I went over the entire piece with 320-grit and dusted it off. Then I charged a rubber (**14**) with super blonde shellac in a 2 lb. cut, and went over all surfaces to apply one coat, in straight strokes with the grain. Again, I brushed the shellac on the stand. Next, I began to body up the case and the stand with the super blonde shellac. To do so, I used a squeeze bottle to squirt a few drops of raw linseed oil onto the surface (**15**). The linseed oil acts as a lubricant, allowing the rubber to skim the surface and preventing previously applied layers of shellac from being disturbed as the new layers are applied. I worked the rubber in circle and figure-eight patterns (**16**), keeping in constant motion to avoid imprinting the surface with the rubber.

Working panel by panel and using light to medium pressure, I recharged the rubber with shellac as necessary and added more oil when I started to feel a drag. I concentrated on the edges, and in doing so, the centers took care of themselves. This time, I used the rubber on the stand as well. For the lid, flap, and spine, I switched to a rubber charged with the previously described shellac/color tint mixture (**17**), allowing me to both tint and body up those areas in one step. On all areas of the piece, I continued this process until the build of the finish was satisfactory and the piece had a consistent sheen.

The last step of French polishing is called "spiriting off" or "stiffing off," and involves the removal of the linseed oil from the finish, which if left can cause problems later in the process. Also, spiriting off actually polishes the shellac, flattening the finish and lessening streaking caused by bodying up. I switched to a clean rubber, charged only with alcohol, and working each panel separately, floated the rubber in straight stokes, with the grain, using slightly less pressure (**18**). As I did so, the oil began to collect on the outer cloth of the rubber, and periodically I rinsed the outer cloth out in the alcohol, squeezed it out, and re-wrapped the rubber. Each time I charged the wadding with less and less alcohol, finally not charging the wadding at all and working with a rubber wrapped in an outer cloth with only the faintest fumes of alcohol.

With the rubber at this stage, I switched to a circular motion and increased my pressure, further lessening streaks. All the while the sheen became shinier and glossier. I stopped when a consistent high gloss was visible, and after all panels were finished, I let the harpsichord rest for two days (**19**).

For the final look I wanted, I decided to dull the sheen slightly, using #0000 steel wool in the direction of the grain. I also waxed the entire piece with Kiwi Bois paste wax [**Hummer Capital, (800) 552-0052**], which gave a nice luster as well as some added protection for the finish (**20**).

Sean Clarke apprenticed with a finishing firm in London at the age of 15 while simultaneously attending the London College of Furniture. He and his wife Angela now operate Clarke Company in Columbus, Ohio.